Thresholds, coherent structures, and dynamics of critical focusing dispersive systems

Kinetic amplification, nonlinear departure, and conserved internal correlations

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Abstract

We connect sharp critical interpolation with actual nonlinear evolution. For the pure mass-critical biharmonic equation, zero-momentum negative-energy solutions at mass excess $d = (M/M_c)^{4/N} - 1 > 0$ satisfy $\sup_{t<T_+} \|\Delta u(t)\|_2^2 \geq c|E|d^{-5/3}$ for sufficiently small d, in a conservative strong flow with the continuation property. The proof imposes no radial symmetry or spatial moment: quantitative constant-phase control produces a slowly moving core, energy splitting bounds exterior mass, and averaging disjoint annuli retains the favorable exterior kinetic term in the virial identity. Amplitude perturbations of sharp standing waves must consequently make arbitrarily large kinetic excursions as their mass approaches the threshold. For slow traveling waves we prove constrained energy descent without real-profile nondegeneracy. In dimensions one through three, with that nondegeneracy explicitly assumed, a small-velocity real spectral pair produces ancient nonlinear orbits on the exact mass–momentum–energy level and logarithmic orbital exit times. Finally, conserved internal correlations and quantitative polarization give global energy-space bounds beyond scalar critical mass for genuinely mixed states, and force every unbounded critical-mass vector evolution to be exactly scalar throughout its lifespan. Complete supporting proofs are included.

The traveling-wave nonlinear theorem retains one explicit real-sector nondegeneracy assumption. The main amplification theorem does not establish finite-time blowup, and the amplification exponent is not claimed optimal. A primary-source comparison is given below.

The main connection

The guiding question is how critical concentration geometry constrains time evolution. For pure quartic dispersion, energy conservation alone gives $A(t) \geq 2|E|/d$. Our first result forces a substantially larger kinetic scale, $c|E|d^{-5/3}$, somewhere along the trajectory. For data $u_d(0) = (1+d)^{N/8}Q$, where Q has mass M_c, this becomes

$$\sup_{0\leq t<T_+} \frac{A(u_d(t))}{A(u_d(0))} \geq cd^{-2/3}.$$

Thus for each prescribed factor K, sufficiently small nonzero d forces a finite time at which the factor K is exceeded. This result concerns a time-dependent solution, without imposing its eventual form.

The exponent has a concrete source. Near-sharp interpolation controls the part outside a constant complex phase by $O(\sqrt{d})$. Local mass conservation then bounds the speed of a concentration center by the same order. Energy splitting leaves at most $O(d)$ mass in a remote tail. Fourth-order interpolation charges a tail current at the rate $L^{4/3}\mu$, producing $d^{2/3}d = d^{5/3}$. Annulus averaging is what allows this argument to use arbitrary nonradial data.

The other two evolution results concern the same relationship from different sides. Constrained phase adjustment determines how motion changes a coherent wave; the spectral pair then yields genuine nonlinear departure, with all conserved quantities fixed. In a unitary internal system, polarization cannot be chosen anew at each time independently of the conserved correlation operator. That conservation law turns a static improvement of sharp interpolation into an all-time bound and an exact scalar reduction at critical mass.

Organization. Section 1 fixes the conventions and compactness input. Section 2 proves the nonradial kinetic amplification result. The next sections treat constrained wave geometry, nonlinear departure, and internal correlations. Appendices reproduce the density geometry and spectral reduction used by these arguments. The concrete strong H^2 flow proofs cover $N = 1, 2, 3$; statements in other dimensions or for general symbols retain their stated local-flow assumptions.

1 Analytic foundations and conventions

The Fourier transform is unitary, $D = -i\nabla$, and $P(u) = \operatorname{Im} \int \overline{u}\nabla u = \int \xi|\widehat{u}|^2$. The real Hilbert pairing is the real part of the complex L^2 pairing. Throughout the local-power arguments, $N \geq 1$, $m > 1$, $s = m/2$, $r = m/N$, and $p = 2 + 2r$. A principal symbol a is real, even, m-homogeneous, C^{N+2} off zero, and satisfies $c|\xi|^m \leq a(\xi) \leq C|\xi|^m$. Set $A(u) = \int a|\widehat{u}|^2$, $M(u) = \|u\|_2^2$, and $E(u) = A(u)/2 - \|u\|_p^p/p$. The sharp critical mass M_c is defined by

$$\|u\|_p^p \leq \frac{p}{2}\left(\frac{M(u)}{M_c}\right)^r A(u). \tag{1}$$

The traveling-wave convention $u(t,x) = e^{i\omega t}U(x - vt)$ gives $[a(D) + \omega + iv \cdot \nabla]U - |U|^{2r}U = 0$.

Lemma 1.1 (Normalized optimizer compactness)**.** *The inequality (1) is attained. If $M(f_n) \to M_c$, $A(f_n) \to 1$, and $\|f_n\|_p^p \to p/2$, then some translated subsequence converges strongly in H^s to an optimizer. The normalized optimizer set is compact modulo translations and phases.*

Proof. Sharp Sobolev interpolation gives a finite positive best constant, and amplitude and spatial scaling normalize a maximizing sequence as stated. Bounded energy-space sequences that vanish locally in mass converge to zero in L^p: use a bounded-overlap cover by unit balls, local Sobolev interpolation, and $2 < p < 2N/(N - m)$ when $N > m$ (all finite p otherwise). Thus translations give a nonzero weak limit Q and almost-everywhere convergence. Write $M_1 = M(Q) > 0$, $A_1 = A(Q) > 0$, $M_2 = M_c - M_1$, and $A_2 = 1 - A_1$. Quadratic splitting and Brezis–Lieb give

$$M_c^r \leq M_1^r A_1 + M_2^r A_2, \qquad A_1 + A_2 = 1.$$

If both masses are positive, the right side is strictly below M_c^r. Consequently $M_1 = M_c$ and then $A_1 = 1$. Weak convergence plus convergence of both quadratic norms gives strong H^s convergence. The same proof applies to a sequence of optimizers. Constant phase is proved by the density-norm lemma below; this compactness argument itself does not need a positivity-preserving kinetic operator. □

Lemma 1.2 (Weak-profile regularity and Pohozaev identities)**.** *Every weak H^s local-power profile belongs to $L^\infty \cap H^{m+1}$ and satisfies*

$$v \cdot P = 2mE, \qquad \omega M = rA + (2m - p)E. \tag{2}$$

No spatial moment is required.

Proof. Choose Λ so large that $a(\xi) - v \cdot \xi + \Lambda \asymp 1 + |\xi|^m$. The shifted resolvent maps L^q to $H^{m,q}$, $1 < q < \infty$, by the Mihlin multiplier theorem. For $N > m$, the starting exponent is $q_0 = 2N/(N-m)$. Splitting the nonlinearity into $|U| > 1$ and $|U| \le 1$, resolvent Sobolev improvement iterates the reciprocal exponent as $\beta_{j+1} = (1+2r)\beta_j - r$ with $\beta_0 = (1-r)/2$. The ideal iterates $\beta_j = 1/2 - (r/2)(1+2r)^j$ cross the boundedness threshold after finitely many steps. Inputs have exponent greater than one; intermediate exponents may be weakened to avoid endpoints. The linear contribution has the stronger improvement $\beta \mapsto \beta - r$ and is handled separately. When $N = m$ start at any sufficiently large finite exponent; when $N < m$ the initial space already embeds in L^∞. Now the nonlinearity is in L^2, giving $U \in H^m \subset H^1$. The C^1 map $z \mapsto |z|^{2r} z$, with bounded derivative on the range of U, sends U into H^1; one more resolvent application gives H^{m+1}.

Here are the cutoff details for the dilation calculation. Let $f_R(x) = x\chi(x/R)$ with χ compactly supported and equal to one near zero, and let $L_R = f_R \cdot \nabla + \operatorname{div} f_R/2$. The Fourier kernel of $[a(D), L_R]$ is, up to the transform constant,

$$\frac{i}{2}(a(\xi) - a(\eta))(\xi + \eta) \cdot \widehat{f}_R(\xi - \eta).$$

Splitting comparable and noncomparable frequencies gives

$$|a(\xi) - a(\eta)||\xi + \eta| \le C|\xi - \eta|(|\xi| + |\eta|)^m.$$

After division by $\langle \xi \rangle^s \langle \eta \rangle^s$, the convolution kernel is bounded by $C|k|\langle k \rangle^s R^{N+1} |\widehat{f}(Rk)|$, with a uniform L^1 norm for $R \ge 1$. Hence the commutators are uniformly bounded $H^s \to H^{-s}$. Schwartz approximation and the homogeneous symbol identity give $\langle U, [a(D), L_R]U \rangle \to mA(U)$. The drift commutator tends to $-v \cdot P$; its order is one and $s > 1/2$. In the nonlinear pairing integration by parts and dominated convergence give

$$\operatorname{Re} \int |U|^{p-2} \overline{U} L_R U \longrightarrow N(1/2 - 1/p)\|U\|_p^p = \frac{m}{p}\|U\|_p^p.$$

Testing the profile with $L_R U$ therefore gives $mE = v \cdot P/2$. Testing with U gives $A + \omega M - v \cdot P - \|U\|_p^p = 0$; combining these two equations proves (2). □

The same regularity proof holds for an individually fixed nonnegative order-m addition b with the stated Mihlin derivative bounds. Constants need not be uniform in that addition or in v. For nonhomogeneous minimizers, differentiation of their scalar mass-preserving dilation energy is legitimate directly by dominated convergence and minimality; differentiability of the dilation orbit as an H^s-valued map is not being assumed.

2 Forced kinetic amplification at nearly critical mass

We return to the pure mass-critical equation

$$iu_t = \Delta^2 u - |u|^{8/N} u, \qquad p = 2 + 8/N, \qquad M = \|u\|_2^2, \quad A = \|\Delta u\|_2^2, \quad E = \frac{A}{2} - \frac{\|u\|_p^p}{p}.$$

Let M_c be the mass in the sharp interpolation inequality

$$\|f\|_p^p \le \frac{p}{2} \left(\frac{M(f)}{M_c} \right)^{4/N} A(f).$$

Solutions are strong conservative H^2 solutions for which the integrated localized mass and virial identities hold by approximation. No spatial moment or radial symmetry is imposed.

Theorem 2.1 (Quantitative kinetic amplification)**.** *There are $d_0, C > 0$, depending only on the scalar sharp interpolation problem, with the following property. If a forward-global solution has*

$$P = 0, \qquad E = -e < 0, \qquad d = \left(\frac{M}{M_c}\right)^{4/N} - 1 \in (0, d_0), \qquad B = \sup_{t \geq 0} A(u(t)) < \infty,$$

then

$$e \leq C d^{5/3} B. \tag{3}$$

Consequently, in any maximal strong H^2 flow with the usual continuation alternative,

$$\sup_{0 \leq t < T_+} A(u(t)) \geq C^{-1} |E| d^{-5/3}. \tag{4}$$

The latter statement allows finite-time blowup, global unbounded motion, and bounded global motion.

The instantaneous sharp interpolation inequality gives only $A(t) \geq 2|E|/d$. Thus (4) forces a larger kinetic scale, by a factor comparable to $d^{-2/3}$. The result does not supply an upper bound on that scale and therefore does not settle finite-time blowup or growup for all such initial data.

2.1 A quantitative constant-phase estimate

For a nonzero scalar complex field define

$$\mathfrak{d}(f) = 1 - \frac{\|f\|_p^p}{(p/2)(M(f)/M_c)^{4/N} A(f)}.$$

Lemma 2.2 (Distance to a real field)**.** *There is $C < \infty$ such that every nonzero $f \in H^2(\mathbb{R}^N; \mathbb{C})$ admits a constant phase θ for which*

$$\frac{M(\mathrm{Im}(e^{-i\theta} f))}{M(f)} + \frac{A(\mathrm{Im}(e^{-i\theta} f))}{A(f)} \leq C \mathfrak{d}(f). \tag{5}$$

No uniqueness of real optimizers or nondegeneracy of the real linearized operator is assumed.

Proof. Amplitude and critical spatial scaling reduce the proof to $M(f) = M_c$ and $A(f) = 1$. If the result failed, since the left side is at most two, there would be a sequence with $\mathfrak{d}(f_n) \to 0$ for which the ratio of the right defect to the best left side tends to zero. Sharp-optimizer compactness supplies, after translations and phases, $f_n \to Q$ strongly in H^2, for a real normalized optimizer Q. Choose the remaining phase so that $f_n = X_n + iY_n$ and $\langle Q, Y_n \rangle = 0$. Put $r = 4/N$, $\kappa = p/2 = 1 + r$, $a_n = A(Y_n)$ and $y_n = M(Y_n)$. The transverse Taylor expansion, with the Sobolev remainder estimate, gives

$$\|X_n + iY_n\|_p^p = \|X_n\|_p^p + \kappa \int |X_n|^{p-2} Y_n^2 + o(\|Y_n\|_{H^2}^2).$$

For $2 < p \leq 4$ the remainder is bounded by $C\|Y_n\|_p^p$; for $p > 4$ use the sum of the fourth-order mixed term and $C\|Y_n\|_p^p$. Sharp interpolation for X_n yields

$$\|X_n\|_p^p \leq \kappa (1 - y_n/M_c)^r (1 - a_n).$$

Potential form convergence therefore implies

$$\kappa - \|f_n\|_p^p \geq \kappa \langle Y_n, L_- Y_n \rangle - o(\|Y_n\|_{H^2}^2), \qquad L_- = \Delta^2 + r/M_c - |Q|^{p-2}.$$

The sharp phase form has kernel exactly $\mathbb{R}Q$ and is coercive on $Q^\perp$, as follows from density strict convexity and the sharp interpolation inequality. Hence the defect controls $\|Y_n\|_{H^2}^2$, a contradiction. Compactness of normalized optimizers makes the coercivity constant uniform. This proves the lemma. □

This is a real two-component polarization statement. Treating f as a single complex internal component would give a vacuous rank-one assertion and would not prove (5).

2.2 A slowly moving concentration center

Critical scaling allows us to assume $B = 1$. Throughout the proof M then lies in a fixed compact interval about M_c, and

$$0 < 2e/d \le A(t) \le 1. \tag{6}$$

Since

$$\mathfrak{d}(u(t)) = \frac{d - 2e/A(t)}{1 + d} \le d,$$

Lemma 2.2 gives, separately at each time, a phase such that $u = e^{i\theta}(X + iY)$ with

$$\|Y\|_{H^2} \le C\sqrt{d}, \qquad \|X\|_{H^2} \le C. \tag{7}$$

No time derivative of the phase is used.

For a real smooth bounded test function ϕ, integration by parts in the mass identity gives

$$\frac{d}{dt}\int \phi|u|^2 = 2\,\mathrm{Im}\int (2\nabla\phi\cdot\nabla\overline{u} + (\Delta\phi)\overline{u})\Delta u. \tag{8}$$

The local nonlinearity cancels. Substituting (7) proves

$$\left|\frac{d}{dt}\int \phi|u|^2\right| \le C\sqrt{d}(\|\nabla\phi\|_\infty + \|\Delta\phi\|_\infty).$$

In particular, for $R \ge 1$ and fixed b,

$$\left|\frac{d}{dt}\int \chi((x-b)/R)|u|^2\right| \le C\sqrt{d}/R. \tag{9}$$

These statements hold in integrated form for the stated solution class.

Choose $\varepsilon_0 = M_c/32$. For sufficiently small d_0, compactness of sharp near-optimizers and (6) give a radius $R_0 \ge 1$ and a choice $c(t)$ satisfying

$$\int_{|x-c(t)|\le R_0} |u(t,x)|^2\,dx \ge M_c - \varepsilon_0. \tag{10}$$

The radius may depend on e/d; no uniform lower kinetic scale is needed. Take $\chi = 1$ on the unit ball and $\chi = 0$ outside the radius-two ball. If $|c(t) - c(s)| > 4R_0$, the integral in (9), centered at $c(s)$, changes by at least $M_c - 3\varepsilon_0$, provided $M - M_c \le \varepsilon_0$. Thus, for a fixed $\tau > 0$,

$$|t - s| \le \tau R_0/\sqrt{d} \quad\Longrightarrow\quad |c(t) - c(s)| \le 4R_0.$$

Interpolate c linearly at time spacing $\tau R_0/\sqrt{d}$. We obtain one Lipschitz center a satisfying

$$\|a'\|_\infty \le L = C\sqrt{d}, \qquad \sup_t |a(t) - c(t)| \le 8R_0. \tag{11}$$

Crucially, its speed bound is independent of R_0 and e/d. A fixed ball about $a(t)$ retains the core mass in (10).

Lemma 2.3 (Uniform mass outside the core)**.** *For every $\eta > 0$ there is $R_\eta < \infty$ such that*

$$\sup_{t\ge 0}\int_{|x-a(t)|>R_\eta} |u(t,x)|^2\,dx \le M - M_c + \eta. \tag{12}$$

Proof. Otherwise choose t_k and $R_k \to \infty$ violating the inequality. After translation by $a(t_k)$, a subsequence converges weakly in H^2 and strongly locally in L^2 to v. The fixed core gives $M(v) \ge M_c - \varepsilon_0$, while the assumed exterior mass implies $M(v) \le M_c - \eta$. The remainder w_k therefore has limiting mass at most $M - M_c + \varepsilon_0 < M_c$. Quadratic splitting and the Brezis–Lieb identity give

$$-e = E(v) + E(w_k) + o(1).$$

Both terms on the right are nonnegative by sharp interpolation, since both masses are strictly subcritical. This is a contradiction. □

The lemma permits a noncompact mass tail of size $M - M_c$. It establishes neither mass compactness nor kinetic compactness of the full trajectory.

2.3 Negative exterior kinetic energy in the virial derivative

Choose a real radial ϕ_R equal to $|x|^2/2$ on $|x| \le R$ and constant on $|x| \ge 2R$, with

$$D^2\phi_R \le I, \qquad \|D^j\phi_R\|_\infty \le C_j R^{2-j}.$$

One construction prescribes $\phi_R'(r) = rh(r/R)$ with smooth nonincreasing h, equal to one on $[0,1]$ and zero on $[2,\infty)$. Set

$$\begin{aligned} J_R(t) &= 2\,\mathrm{Im}\int \nabla\phi_R(x-a(t))\cdot\overline{u}\nabla u,\\ K_R(t) &= \int_{|x-a(t)|>2R} |D^2u|^2,\\ H_R(t) &= \int_{R/2<|x-a(t)|<2R} |D^2u|^2. \end{aligned}$$

We use the full Hessian in these tails. Its total squared norm equals $\|\Delta u\|_2^2$, but the two densities are not pointwise identical.

The exact localized quartic identity is

$$\begin{aligned} J_R' = {}& 8\,\mathrm{Re}\int \partial_{jk}\phi_R\,\partial_{ij}\overline{u}\,\partial_{ik}u - 4\,\mathrm{Re}\int \partial_{jk}\Delta\phi_R\,\partial_j\overline{u}\,\partial_k u\\ & - 2\int \Delta^2\phi_R|\nabla u|^2 + \int \Delta^3\phi_R|u|^2 - \frac{16}{Np}\int \Delta\phi_R|u|^p - 2a'\cdot\int D^2\phi_R\,\mathrm{Im}(\overline{u}\nabla u). \end{aligned}$$

The cutoff is centered at $a(t)$. Relative to the global kinetic term $8A$, the first term has a nonpositive defect everywhere, and its defect is exactly $-8|D^2u|^2$ outside $2R$. Higher cutoff derivatives contribute $O(R^{-2})$. Subtracting the total momentum from the last term and using $P = 0$ therefore gives

$$J_R' \le -16e - 8K_R + C\int_{|x-a|>R} |u|^p + CR^{-2} + CL\int_{|x-a|>R} |u||\nabla u|. \tag{13}$$

Also $|J_R| \le CR$.

Fix small $\eta > 0$, put $\mu = M - M_c + \eta$, and take $R \ge 2R_\eta$. For a cutoff ζ_R vanishing inside $R/2$ and equal to one outside R, Lemma 2.3 and the product rule give

$$M(\zeta_R u) \le \mu, \qquad A(\zeta_R u) \le C(K_R + H_R + R^{-2}).$$

Sharp interpolation and $\|\nabla f\|_2^2 \le \|f\|_2\|\Delta f\|_2$ imply

$$\begin{aligned} \int_{|x-a|>R} |u|^p &\le C\mu^{4/N}(K_R + H_R + R^{-2}),\\ \int_{|x-a|>R} |u||\nabla u| &\le C\mu^{3/4}((K_R + H_R)^{1/4} + R^{-1/2}). \end{aligned}$$

Choose d_0 and η so small that $C\mu^{4/N} \le 1$. Absorb the terms containing K_R into its negative coefficient in (13); for the current term use Young's inequality. We obtain

$$J_R' \le -16e + C\mu^{4/N}H_R + CL\mu^{3/4}H_R^{1/4} + CL^{4/3}\mu + o_R(1), \tag{14}$$

where $o_R(1) \to 0$ uniformly in time. No favorable lower bound on the kinetic defect in the transition annulus was used.

2.4 Averaging disjoint annuli

Fix a positive integer K and $R_* \geq 2R_\eta$. Let $R_j = 8^j R_*$ for $0 \leq j < K$. Their transition annuli are disjoint about the same center, so

$$\sum_{j=0}^{K-1} H_{R_j}(t) \leq A(t) \leq 1.$$

For each T choose an index whose time-averaged H_{R_j} is at most K^{-1}. Integrate (14), use Jensen's inequality for the fourth root, and bound both endpoint values of J_{R_j}:

$$16e \leq \frac{2C8^{K-1}R_*}{T} + \frac{C\mu^{4/N}}{K} + CL\mu^{3/4}K^{-1/4} + CL^{4/3}\mu + \sup_{R\geq R_*} |o_R(1)|.$$

First let $T \to \infty$ with K, R_* fixed, then $K \to \infty$, then $R_* \to \infty$, and finally $\eta \downarrow 0$. This proves

$$e \leq CL^{4/3}(M - M_c).$$

The radius is fixed during each time integration. By (11), $L \leq C\sqrt{d}$, and $M - M_c \leq Cd$. Thus $e \leq Cd^{5/3}$ when $B = 1$. Undoing critical scaling proves Theorem 2.1.

2.5 Finite-time excursions from every sharp standing wave

Corollary 2.4 (Standing-wave instability without real-sector nondegeneracy)**.** *Let Q be any real sharp optimizer normalized by $M(Q) = M_c$, with its associated standing-wave frequency. Assume a strong H^2 local flow and the continuation alternative. For*

$$u_d(0) = (1 + d)^{N/8}Q, \qquad d > 0,$$

and every prescribed $K > 1$, all sufficiently small d admit a finite forward time $t_d < T_+(d)$ for which

$$A(u_d(t_d)) > KA(u_d(0)).$$

In particular the standing wave is orbitally unstable in H^2 modulo constant phase and translations. No radial or L_+ nondegeneracy hypothesis is needed.

Proof. These initial data have $P = 0$, $M = (1 + d)^{N/4}M_c$ and $E = -dA(u_d(0))/2$. Theorem 2.1 gives

$$\sup_{0\leq t<T_+} \frac{A(u_d(t))}{A(u_d(0))} \geq (2C)^{-1}d^{-2/3}.$$

For small d this is strictly greater than K, so a finite time has the claimed property. Since $u_d(0) \to Q$ in H^2, take, for example, $K = 4$ and use

$$\inf_{\theta,y} \|u_d(t_d) - e^{i\theta}Q(\cdot - y)\|_{H^2} \geq \|\Delta u_d(t_d)\|_2 - \|\Delta Q\|_2 \geq \|\Delta Q\|_2.$$

This proves orbital instability. □

For $N = 1, 2, 3$ the required strong local flow follows from the H^2 algebra and composition estimates, with conservation and the integrated identities obtained by smooth approximation. In larger dimensions the statement uses the appropriate strong energy-space local theory. The conclusion is a finite-time excursion for the perturbed data; its proof does not identify which of finite blowup, global unbounded evolution, or bounded large motion occurs.

3 The energy geometry at fixed mass and momentum

The spectral reduction for a specified smooth branch requires control of the real linearized kernel. The variational assertion below does not: the imaginary response alone forces a negative constrained energy curvature when the homogeneous order exceeds two. Every component of momentum is preserved, and no differentiability of the dilation group on the individual profile is assumed.

Let $s = m/2$, $\alpha = 2m/N$, and let a be a positive, real-even, elliptic homogeneous symbol of degree $m > 1$. Write

$$E(u) = \frac{1}{2}A(u) - \frac{1}{\alpha+2}\|u\|_{\alpha+2}^{\alpha+2}, \quad M(u) = \|u\|_2^2, \quad P(u) = \operatorname{Im}\int \overline{u}\nabla u.$$

Let Q be a real sharp ground profile at frequency one. The scalar phase-coercivity theorem gives

$$L_- = a(D) + 1 - |Q|^\alpha \geq 0, \qquad \ker L_- = \mathbb{R}Q,$$

with coercivity on $H^s \cap Q^\perp$. Define

$$Z_j = L_-^{-1}\partial_j Q, \qquad T_{ij} = \langle \partial_i Q, Z_j \rangle. \tag{15}$$

The inverse and pairing can be understood in energy-space form sense. In particular T is positive definite.

Theorem 3.1 (Constrained curvature without a profile branch)**.** *Suppose $v_n \to 0$, $v_n \neq 0$, and exact profiles satisfy*

$$(a(D) + 1 + iv_n \cdot \nabla)U_n - |U_n|^\alpha U_n = 0, \qquad U_n \longrightarrow Q \quad \textit{strongly in } H^s,$$

after phase and translation. For every sufficiently large n there is a continuous H^s curve $V_n(\lambda)$, defined near $\lambda = 1$, with

$$V_n(1) = U_n, \qquad M(V_n(\lambda)) = M(U_n), \qquad P(V_n(\lambda)) = P(U_n).$$

Its scalar energy is C^2 and obeys

$$\frac{d}{d\lambda}E(V_n(\lambda))\Big|_1 = 0, \qquad \frac{d^2}{d\lambda^2}E(V_n(\lambda))\Big|_1 = (2-m)v_n^T T v_n + o(|v_n|^2). \tag{16}$$

For $m > 2$ the profile is therefore not a local energy minimum at its fixed mass and momentum: both sides of this curve have strictly lower energy. The statement assumes neither a differentiable family of profiles nor nondegeneracy of the real block L_+.

Proof. We suppress the sequence index and keep errors relative to its limit. Choose the phase so that $U = X + iY$ and $\langle Y, Q\rangle = 0$. The imaginary profile equation is

$$L_- Y = -v \cdot \nabla X + (|U|^\alpha - |Q|^\alpha)Y.$$

Put $p = \alpha + 2$. Since $H^s \hookrightarrow L^p$, the potential $|U|^\alpha - |Q|^\alpha$ tends to zero as a quadratic form on H^s: this follows from Hölder and its convergence in $L^{p/\alpha}$. The derivative maps H^s continuously to H^{-s} because $2s = m > 1$. Project onto $Q^\perp$ and apply phase coercivity. First $\|Y\|_{H^s} = O(|v|)$; substituting this bound back into the equation gives

$$Y = -\sum_j v_j Z_j + o(|v|) \quad \text{in } H^s. \tag{17}$$

No estimate for a derivative of the real profile is needed. The bilinear momentum form is continuous on H^s, and $P_j(U) = -2\langle Y, \partial_j X\rangle$. Hence

$$P(U) = 2Tv + o(|v|). \tag{18}$$

The positivity of T follows from phase coercivity and the fact that a nonzero L^2 function cannot have a vanishing nonzero directional derivative.

For $c = (c_0, c_1, \dots, c_N) \in \mathbb{R}^{N+1}$ set

$$W(c) = (1 + c_0)U + i \sum_{j=1}^{N} c_j Z_j.$$

The Jacobian at $c = 0$ of $c \mapsto (M(W(c)), P(W(c)))$ is

$$\begin{pmatrix} 2M(U) & O(|v|) \\ 2P(U) & -2T + o(1) \end{pmatrix}.$$

It is invertible for sufficiently small velocity. The finite-dimensional implicit-function theorem supplies a smooth $c(\lambda)$ with $c(1) = 0$ and

$$M(W(\lambda)) = M(U), \qquad P(W(\lambda)) = \lambda^{-1} P(U). \tag{19}$$

Here and below $W(\lambda)$ denotes $W(c(\lambda))$. Differentiating these equations and using (18) gives

$$c_j'(1) = v_j + o(|v|), \qquad c_0'(1) = O(|v|^2), \qquad h := W'(1) = i \sum_j v_j Z_j + o(|v|) \quad \text{in } H^s.$$

The amplitude estimate uses $\langle U, iZ_j \rangle = \langle Y, Z_j \rangle = O(|v|)$.

Define

$$V(\lambda, x) = \lambda^{N/2} W(\lambda, \lambda x).$$

Critical homogeneity gives the exact identities

$$M(V(\lambda)) = M(U), \quad P(V(\lambda)) = P(U), \quad E(V(\lambda)) = \lambda^m E(W(\lambda)). \tag{20}$$

Strong continuity of dilation gives a continuous H^s curve. Although that curve need not be differentiable in H^s, its scalar energy is C^2 by the last identity. This is the reason that no weighted moment or dilation-domain assumption enters the theorem.

Let $\mathcal{L}_U$ be the real Hessian of $S = E + M/2 - v \cdot P/2$, and put $q_U(h) = \langle h, \mathcal{L}_U h \rangle$ in form notation. Because $S'(U) = 0$, differentiating (19) yields

$$\frac{d}{d\lambda} E(W(\lambda))\Big|_1 = -\frac{1}{2} v \cdot P(U), \qquad \frac{d^2}{d\lambda^2} E(W(\lambda))\Big|_1 = q_U(h) + v \cdot P(U).$$

Using the exact profile identity $v \cdot P = 2mE$ in (20), we conclude that

$$E(V)'(1) = 0, \qquad E(V)''(1) = q_U(h) - \frac{m-1}{2} v \cdot P(U). \tag{21}$$

The energy is a C^2 functional on H^s, since $p > 2$ and $H^s \hookrightarrow L^p$. Its Hessian forms therefore converge on bounded sets as $U \to Q$, $v \to 0$. The imaginary block at Q is L_-, so

$$q_U(h) = q_Q\Big(i \sum_j v_j Z_j\Big) + o(|v|^2) = v^T T v + o(|v|^2).$$

Together with (18) and (21), this proves (16). The sign for $m > 2$ follows from positive definiteness of T. □

Corollary 3.2 (All sufficiently slow action ground waves)**.** *At frequency one, define*

$$K_v(u) = A(u) + M(u) - v \cdot P(u), \qquad \kappa_v = \inf_{u \neq 0} \frac{K_v(u)}{\|u\|_p^2}, \qquad p = 2 + \frac{2m}{N}.$$

For sufficiently small $|v|$, this infimum is attained. Normalize each minimizer U by $K_v(U) = \|U\|_p^p$; these are precisely the action minimizers on the nonzero Nehari manifold. If $m > 2$, then every such ground wave at every sufficiently small nonzero velocity is not a local energy minimum at fixed mass and momentum. This conclusion requires no uniqueness, differentiable branch, or real-profile nondegeneracy.

Proof. Ellipticity and Hölder in Fourier variables give

$$|P(u)| \le CA(u)^{1/m} M(u)^{1-1/m} \le C'(A(u) + M(u)).$$

Consequently K_v is uniformly equivalent to the H^s norm squared for sufficiently small $|v|$. The exponent $p > 2$ is strictly below the energy-space Sobolev endpoint, including when $s \ge N/2$. For completeness, normalize a minimizing sequence by $\|u_n\|_p = 1$. It is bounded in H^s and cannot vanish locally, by the standard localized Sobolev interpolation argument. After a translation it has a nonzero weak limit u. Local compact embedding and the Brezis–Lieb identity split the pth power norm into c and $1 - c$, where $c = \|u\|_p^p > 0$. Weak Hilbert-space orthogonality splits K_v. Thus

$$\kappa_v \ge \kappa_v\big(c^{2/p} + (1-c)^{2/p}\big).$$

Since $2/p < 1$, this forces $c = 1$. Norm convergence and the coercivity of K_v give strong H^s convergence and attainment. Rescaling a minimizer to the Nehari manifold gives the profile equation and

$$K_v(U) = \|U\|_p^p = \kappa_v^{p/(p-2)}. \tag{22}$$

On that manifold $S_v(U) = (1/2 - 1/p)\|U\|_p^p$, proving the asserted minimizing equivalence.

The Fourier bound also gives $\kappa_v \to \kappa_0$ as $v \to 0$. Let $v_n \to 0$ and let U_n be any Nehari-normalized minimizers. They are bounded in H^s by (22), and $K_0(U_n) - K_{v_n}(U_n) = v_n \cdot P(U_n) \to 0$. Therefore their normalized versions form a minimizing sequence for κ_0. The preceding compactness proof gives, after translations and passing to a subsequence, $U_n \to Q$ strongly in H^s, where Q is a frequency-one ground profile.

We verify that this limiting profile is one of the sharp critical optimizers used in Theorem 3.1. Write $r = m/N$ and let the sharp critical inequality be

$$\|u\|_p^p \le C_* M(u)^r A(u).$$

For fixed $A + M$, the product $M^r A$ is maximal at $M = rA$. A critical dilation of a sharp optimizer attains that ratio, so it also attains κ_0. Conversely, attainment of κ_0 forces equality in both inequalities, hence sharp critical optimization and $M = rA$. The density phase-rigidity theorem makes this optimizer real after a constant phase. Nehari normalization then gives $\|Q\|_p^p = (1+r)A(Q)$, and equality in $\|Q\|_p^p \le (1+r)(M(Q)/M_c)^r A(Q)$ gives $M(Q) = M_c$.

If the asserted uniform small-velocity conclusion failed, there would be a sequence of nonzero $v_n \to 0$ and ground waves lacking a negative fixed-charge curve. The compactness just proved produces a real limiting Q, to which Theorem 3.1 applies. Its strictly negative curvature contradicts that failure. No uniform choice of a differentiable family was used. □

Why the phase correction matters. The linear boost

$$V_b(\lambda, x) = e^{i(1-\lambda)k\cdot x}\lambda^{N/2} U(\lambda x), \qquad k = \frac{P(U)}{M(U)},$$

also fixes both charges. However its curvature, when a has the homogeneous C^2 derivative bounds and $m > 2$, is

$$E(V_b)''(1) = \frac{1}{2}\left(k^T H_U k - (m-1)v \cdot P(U)\right), \qquad H_U = \int D^2 a(\xi)|\widehat{U}(\xi)|^2\, d\xi. \tag{23}$$

Indeed the kinetic term is $\int a(k+\lambda(\xi-k))|\widehat{U}(\xi)|^2\,d\xi$. Euler's identities and the profile boost identity $\int \nabla a|\widehat{U}|^2 = vM$ give (23). Near Q its leading term is

$$\frac{2}{M(Q)^2}v^T T H_Q T v - (m-1)v^T T v.$$

Phase positivity alone gives no sign for this expression. By contrast the optimal imaginary correction minimizes $\langle\psi, L_-\psi\rangle$ under $-2\langle\partial_j Q,\psi\rangle = -P_j$. Its minimum is $P^T T^{-1} P/4$, attained at $\psi = \frac{1}{2}\sum_j (T^{-1}P)_j Z_j$. This is the correction used to leading order in the theorem.

Remark 3.3 (Energetic and dynamical conclusions). The theorem gives energy-decreasing perturbations at precisely the same conserved charges as the wave. It does not alone prove that their solutions escape a neighborhood of the symmetry orbit. In particular, Morse index one does not remove an uncontrolled extra kernel: a negative direction can belong to a zero Jordan chain. Nonlinear or spectral instability requires the additional dynamical argument stated in its own theorem. For $1 < m < 2$ the displayed curve instead has positive leading curvature, which also does not establish stability.

4 Nonlinear departure on the exact invariant level

We now turn the real spectral pair into an actual nonlinear trajectory. The equation throughout this section is the pure mass-critical quartic model

$$iu_t = \Delta^2 u - |u|^{8/N}u, \qquad N \in \{1,2,3\}, \qquad X = H^2(\mathbb{R}^N;\mathbb{C}). \tag{24}$$

The space X is regarded as real. In this concrete phase space, the branch, weighted regularity, and complementary gap used in the spectral theorem in Appendix B follow from real-profile nondegeneracy. That remaining assumption is stated explicitly below. The form decomposition is the one-negative-direction instance of the Hamiltonian trichotomy developed by Lin and Zeng [7]; the estimates needed here are proved directly.

Theorem 4.1 (Ancient departure and nonlinear orbital instability). *Let Q be a real sharp ground profile at frequency one. Assume*

$$\ker\left(\Delta^2 + 1 - \left(1+\frac{8}{N}\right)|Q|^{8/N}\right) = \operatorname{span}_{\mathbb{R}}\{\partial_1 Q,\dots,\partial_N Q\}. \tag{25}$$

Then there is a local C^2 profile family $U_{\omega,v}$ through Q, unique up to phase and translation near Q. Put $U_v = U_{1,v}$. For every sufficiently small fixed $v \neq 0$, put

$$\mathcal{O}_v = \{e^{i\theta}U_v(\cdot - y) : \theta \in \mathbb{R},\ y \in \mathbb{R}^N\}.$$

There is a nonstationary solution in the co-moving, co-rotating frame, defined for $t \le 0$, of the form

$$w_v(t) = U_v + \varepsilon_v e^{\lambda_v t}Y_{+,v} + O_v(e^{(1+\beta)\lambda_v t}), \qquad t \to -\infty, \tag{26}$$

where $\|Y_{+,v}\|_{H^2} = 1$, $J\mathcal{L}_v Y_{+,v} = \lambda_v Y_{+,v}$, $\varepsilon_v > 0$, and

$$\beta = \begin{cases} 1, & N = 1,2, \\ 2/3, & N = 3. \end{cases}$$

At every time this solution has exactly the same mass, momentum, and energy as U_v.

Consequently, there are $\rho_v > 0$ and data $u_{\delta,v}$ satisfying

$$M(u_{\delta,v}) = M(U_v), \quad P(u_{\delta,v}) = P(U_v), \quad E(u_{\delta,v}) = E(U_v),$$
$$u_{\delta,v} = U_v + \delta Y_{+,v} + O_v(\delta^{1+\beta}) \quad \text{in } H^2,$$

whose first exit time from the orbital distance ρ_v obeys

$$\tau_{\delta,v} = \lambda_v^{-1} \log(1/\delta) + O_v(1) \qquad (\delta \downarrow 0). \tag{27}$$

The spectral calculation gives

$$\lambda_v = \sqrt{\frac{2v^T T v}{\mathcal{D}_Q}} + o(|v|), \tag{28}$$

where

$$T_{ij} = \langle \partial_i Q, L_-^{-1} \partial_j Q \rangle, \qquad c_j = \langle \Lambda Q, L_-^{-1} \partial_j Q \rangle,$$
$$\mathcal{D}_Q = \langle \Lambda Q, L_-^{-1} \Lambda Q \rangle - c^T T^{-1} c.$$

In particular, $e^{it} U_v(x - vt)$ is orbitally unstable in H^2 even on its exact joint mass–momentum–energy level.

4.1 Local flow and the nonlinear remainder

Set $F(z) = |z|^{8/N} z$, with $\mathbb{C}$ identified with $\mathbb{R}^2$. Since $H^2 \hookrightarrow L^\infty \cap W^{1,4}$ for $N \le 3$ and H^2 is an algebra, $F : X \to X$ is locally $C^{1,\beta}$ with the exponent in the theorem. Here is a direct check of the regularity that is needed. Twice differentiating $DF(w)h$ produces sums of

$$DF(w)\nabla^2 h, \qquad D^2F(w)[\nabla w, \nabla h],$$
$$D^2F(w)[\nabla^2 w, h], \qquad D^3F(w)[\nabla w, \nabla w, h].$$

The products lie in L^2 by the stated embeddings. On bounded subsets of $\mathbb{R}^2$, D^3F is Lipschitz in dimensions one and two, and 2/3-Hölder in dimension three. Subtracting these expressions at w_1, w_2 therefore gives

$$\|DF(w_1) - DF(w_2)\|_{\mathcal{B}(X)} \le C_R \|w_1 - w_2\|_X^\beta$$

for $\|w_i\|_X \le R$ and $\|w_1 - w_2\|_X \le 1$. The lower derivative terms obey the same bound. Smooth approximation justifies the weak chain rules. It follows that, near a fixed U,

$$\|F(U + h) - F(U) - DF(U)h\|_X \le C\|h\|_X^{1+\beta}, \tag{29}$$
$$\|R(h_1) - R(h_2)\|_X \le C(\|h_1\|_X^\beta + \|h_2\|_X^\beta)\|h_1 - h_2\|_X, \tag{30}$$

where $R(h) = i[F(U + h) - F(U) - DF(U)h]$.

The free group $e^{-it\Delta^2}$ is unitary on X. A contraction argument gives local well-posedness and continuation while the H^2 norm is bounded. Mass, momentum, and energy are conserved. For a direct justification, insert a selfadjoint Fourier cutoff P_n into the Hamiltonian nonlinearity $P_n F(P_n u)$, differentiate the conserved quantities for this regularized flow, and pass to the H^2 solution uniformly on compact existence intervals. The cutoff commutes with phase and translation actions. Thus no Strichartz or spectral-mapping hypothesis is required for the local nonlinear flow.

For

$$w(t, x) = e^{-it} u(t, x + vt) = U_v + h(t, x)$$

the perturbation equation is

$$h_t = \mathcal{A}_v h + R_v(h), \qquad \mathcal{A}_v = J\mathcal{L}_v, \quad J = -i. \tag{31}$$

The phase and translation transformations are H^2 isometries and preserve all three conserved quantities.

4.2 Why the other spectral assumptions follow here

Lemma 4.2 (Local branch and complementary gap)**.** *For a real sharp quartic ground profile satisfying (25), all the other assumptions of the small-velocity theorem in Appendix B hold. In particular, $\Lambda Q \in H^4$, the local profile branch is C^2 into H^4, and its only Hessian zero modes are phase and translations.*

Proof. Sharp phase coercivity gives $L_- \geq 0$, $\ker L_- = \mathbb{R}Q$, and an inverse on $Q^\perp$. The nonlinearity calculation above and $(\Delta^2 + 1)Q = F(Q)$ give $Q \in H^6$. Therefore $V = |Q|^{8/N} \in W^{2,\infty}$ and $V(x) \to 0$ as $|x| \to \infty$. We first justify the scaling derivative without assuming a moment.

Choose smooth bounded truncations ϕ_R of $a\langle x\rangle$, with $\phi_R \to a\langle x\rangle$ pointwise, $\phi_R \leq Ca\langle x\rangle$, and $|\nabla\phi_R| + |\Delta\phi_R| \leq Ca$ uniformly in R. Test $(\Delta^2 + 1 - V)Q = 0$ with $e^{2\phi_R}Q$ and put $w_R = e^{\phi_R}Q$. The exact real identity

$$\mathrm{Re}\langle \Delta^2 Q, e^{2\phi_R}Q\rangle = \|\Delta w_R + |\nabla\phi_R|^2 w_R\|_2^2 - \|2\nabla\phi_R\cdot\nabla w_R + (\Delta\phi_R)w_R\|_2^2$$

shows that the difference from $\|\Delta w_R\|_2^2$ is bounded by $Ca^2\|w_R\|_{H^2}^2$ for $a \leq 1$. Choose a fixed ball outside which $V \leq 1/4$. Inside the ball, $\int V|w_R|^2$ is uniformly bounded because the weights there are bounded independently of R. Taking $a > 0$ small absorbs both the exterior potential and the Ca^2 error. Thus w_R is uniformly bounded in H^2, and passage to the limit gives $w = e^{a\langle x\rangle}Q \in H^2$.

The conjugated equation

$$e^{a\langle x\rangle}(\Delta^2 + 1)e^{-a\langle x\rangle}w = Vw$$

has smooth bounded coefficients in its lower-order terms, of order at most three. Since $V \in W^{2,\infty}$, elementary elliptic bootstrapping improves w successively from H^2 to H^3, H^4, H^5, H^6. At the first step, the third-order terms belong to H^{-1}; at the subsequent steps they belong to L^2, H^1, H^2, respectively. Consequently $\Lambda Q = NQ/2 + x\cdot\nabla Q \in H^4$, and $L_+\Lambda Q = -4Q$ follows by differentiating the scaled equation.

For the branch, set

$$\mathcal{F}(U,\omega,v) = \Delta^2 U + \omega U + iv\cdot\nabla U - F(U).$$

This map is C^2 from $H^4 \times \mathbb{R}^{1+N}$ to L^2. Let $g_0 = iQ$, $g_j = \partial_j Q$, and augment the profile equation by

$$\mathcal{F}(U,\omega,v) = \sum_{j=0}^{N} b_j g_j, \qquad \langle U - Q, g_j\rangle_{\mathbb{R}} = 0.$$

Its derivative with respect to (U, b) is invertible: the real selfadjoint Hessian at Q is Fredholm, its kernel is exactly the span of the g_j, and it has a bounded complementary inverse from L^2 to H^4. The implicit function theorem gives a C^2 augmented branch. Phase and translation invariance imply

$$\langle \mathcal{F}(U,\omega,v), iU\rangle_{\mathbb{R}} = 0, \qquad \langle \mathcal{F}(U,\omega,v), \partial_k U\rangle_{\mathbb{R}} = 0.$$

The pairing matrix between the fixed g_j and these moving symmetry vectors is close to the invertible Gram matrix at Q. Hence $b = 0$, and the branch consists of exact profiles.

The gauge determines profiles uniquely near Q up to phase and translation. Mass-preserving scaling maps parameters to $(\kappa^4\omega, \kappa^3 v)$; after returning to the gauge, uniqueness shows that the branch is scaling-closed. Its reduced action is C^2 and has the usual charge derivatives. Differentiation at $(1, 0)$ gives $D_{v_j}U = -iL_-^{-1}\partial_j Q$ in the fixed gauge.

Finally, the Hessians have common domain H^4. The potentials converge in L^∞, and $iv\cdot\nabla$ is small from H^4 to L^2. Norm-resolvent convergence therefore preserves the rank $N + 1$ of a small isolated spectral cluster about zero. The $N + 1$ independent vectors $iU, \partial_j U$ fill that cluster. They belong to the operator domain by the profile equation and elliptic bootstrapping. Thus there are no further zero or near-zero eigenvalues, and the inverse on the kernel complement, extended by zero, is norm-continuous on L^2. This verifies the complementary-gap assumption rather than adding it. □

4.3 The linear group bound from one negative direction

Lemma 4.3. *For each fixed sufficiently small $v \neq 0$,*

$$n_-(\mathcal{L}_v) = 1, \qquad \|e^{t\mathcal{A}_v}\|_{\mathcal{B}(H^2)} \leq C_v e^{\lambda_v t} \quad (t \geq 0).$$

On an invariant complement to the two real eigenvectors the group has at most linear growth.

Proof. At $v = 0$, sharp critical optimality gives nonnegativity of L_+ on $Q^\perp$ and

$$\langle Q, L_+ Q\rangle = -\frac{8}{N}\|Q\|_{2+8/N}^{2+8/N} < 0.$$

Since $L_- \geq 0$, the full Hessian has exactly one negative direction. For small v, the free operator

$$H_v = \Delta^2 + 1 + iv\cdot\nabla$$

has symbol $|\xi|^4 + 1 - v\cdot\xi$ uniformly comparable to $1 + |\xi|^4$. The potential $DF(U_v)$ is a decaying bounded real symmetric matrix and converges in L^∞ to $DF(Q)$. The drift is small from H^4 to L^2. This gives norm-resolvent convergence with a common lower bound and a uniformly positive essential spectrum. The isolated zero cluster contains exactly the prescribed $N + 1$ symmetry modes. Hence $n_-(\mathcal{L}_v) = 1$ for small v. This concrete perturbation argument excludes negative spectrum arriving from infinity, as well as from zero.

Fix v and omit its subscript. The profile equation and $F(U) \in H^2$ give $U \in H^6$. The matrix multiplier $DF(U)$ is bounded on H^2. The free operator $-iH_v$ generates an isometric group on H^2; its bounded real-linear perturbation $iDF(U)$ therefore generates a C^0 group there. The L^2 eigenvectors of $\mathcal{A}$ lie in H^4, and their eigenvalue equations improve them to H^6.

Let $\mathcal{A}Y_\pm = \pm\lambda Y_\pm$, $\lambda > 0$, and write $B(f, g) = \langle \mathcal{L}f, g\rangle$ for the real form. The identity

$$B(\mathcal{A}f, g) + B(f, \mathcal{A}g) = 0$$

gives $B(Y_\pm, Y_\pm) = 0$. Furthermore $b = B(Y_+, Y_-) \neq 0$: otherwise their span would be a two-dimensional totally isotropic space modulo $\ker\mathcal{L}$, impossible for a form of negative index one. Their span is disjoint from the kernel, as is seen by applying $\mathcal{A}$ and $\mathcal{A}^2$ to any proposed relation. Hence

$$W = \operatorname{span}_{\mathbb{R}}\{Y_+, Y_-\}, \qquad Z = W^{\perp_B}, \qquad X = W \oplus Z$$

is an invariant decomposition with bounded projection

$$\Pi_W h = \frac{B(h, Y_-)}{b}Y_+ + \frac{B(h, Y_+)}{b}Y_-.$$

The form is nonnegative on Z, because its sole negative direction is already in W. Its null space in Z is exactly $K = \ker\mathcal{L}$: if $B(z, z) = 0$, nonnegativity implies $B(z, Z) = 0$, while the definition gives $B(z, W) = 0$.

Let Π_K be the L^2 orthogonal projection onto K. Compactness of the decaying potential relative to the positive free form gives

$$B(z, z) \geq c_v\|(I - \Pi_K)z\|_{H^2}^2, \qquad z \in Z. \tag{32}$$

Indeed, a violating sequence in $Z \cap K^\perp$, normalized in H^2, has a weak limit. Weak lower semicontinuity of the free form and compactness of the potential force its form value to be nonpositive. The preceding null-space statement makes that limit zero. Compactness then makes the potential term vanish, contradicting coercivity of the free form and the normalization.

The form $B(e^{t\mathcal{A}}z, e^{t\mathcal{A}}z)$ is conserved, first on the generator domain and then by density. It controls the component orthogonal to K through (32). For an L^2 orthonormal basis g_j of K, write $h = z + k$ with $k = \Pi_K h$. In the H^{-2}–H^2 pairing,

$$\frac{d}{dt}\langle h, g_j\rangle = -\langle \mathcal{L}z, Jg_j\rangle, \qquad \left|\frac{d}{dt}\langle h, g_j\rangle\right| \leq C_v\|z\|_{H^2}.$$

This proves

$$\|e^{t\mathcal{A}}|_Z\|_{\mathcal{B}(H^2)} \le C_v(1+|t|).$$

The group on W is exactly $e^{\pm\lambda t}$. Absorbing $1+t$ into a λ-dependent multiple of $e^{\lambda t}$ proves the lemma. □

4.4 An ancient nonlinear orbit and exact conservation

Proof of Theorem 4.1. Lemma 4.2 verifies the hypotheses of the calculation in Appendix B and hence supplies λ_v and $Y_{+,v}$. Fix v and suppress it. Normalize $\|Y_+\|_X = 1$. On continuous functions $h : (-\infty, 0] \to X$ use the norm

$$\|h\|_* = \sup_{t\le 0} e^{-\lambda t}\|h(t)\|_X.$$

For small $\varepsilon > 0$, consider on the ball $\|h\|_* \le 2\varepsilon$ the map

$$\Phi(h)(t) = \varepsilon e^{\lambda t}Y_+ + \int_{-\infty}^{t} e^{(t-s)\mathcal{A}}R(h(s))\,ds. \tag{33}$$

The integral converges in X, since Lemma 4.3 and (29) give

$$\|\Phi(h)(t) - \varepsilon e^{\lambda t}Y_+\|_X \le \frac{C_v C_R 2^{1+\beta}}{\beta\lambda}\varepsilon^{1+\beta}e^{(1+\beta)\lambda t}. \tag{34}$$

By (30), its weighted Lipschitz constant is at most $C_v'\varepsilon^\beta/(\beta\lambda)$. Choose ε small enough for invariance of the ball and a contraction constant below $1/2$. The fixed point is a nonlinear mild solution on every finite subinterval: split the integral in (33) at a finite initial time and use the group law. It satisfies (26) and converges strongly to U backward in time. Conservation and continuity of M, P, E therefore imply

$$(M,P,E)(U+h(t)) = (M,P,E)(U), \qquad t \le 0. \tag{35}$$

It remains to establish departure from the entire symmetry orbit. The map $(\theta, y) \mapsto e^{i\theta}U(\cdot - y)$ is C^2 into H^2, with injective differential and tangent K. It has no nontrivial stabilizer: a nonzero phase-periodic translation contradicts $U \in L^2$. Nor can translations tending to infinity approach U in L^2. Thus every orbit point sufficiently near U belongs to this local chart. Put

$$\eta_v = \operatorname{dist}_{H^2}(Y_+, K) > 0.$$

It is positive because $\mathcal{A}$ annihilates K and has nonzero eigenvalue on Y_+. Taylor expansion of the chart gives, for small $a > 0$,

$$\operatorname{dist}_{H^2}(U + aY_+, \mathcal{O}_v) \ge \frac{\eta_v}{2}a. \tag{36}$$

Indeed, any competing orbit point with comparable distance lies in the chart with parameters $O(a)$; projecting its expansion off K proves the inequality. No existence of a global nearest orbit point is needed.

Shrink ε_v further so that (36) holds at $a = \varepsilon_v$, and the right side of (34) at $t = 0$ is at most $\eta_v\varepsilon_v/4$. Then

$$\operatorname{dist}_{H^2}(U + h(0), \mathcal{O}_v) \ge \eta_v\varepsilon_v/4.$$

Set $\rho_v = \eta_v\varepsilon_v/8$. For $0 < \delta < \varepsilon_v$ let $t_\delta = \lambda^{-1}\log(\delta/\varepsilon_v)$ and $u_{\delta,v} = U + h(t_\delta)$. Its expansion and exact invariant values follow from (34) and (35). Its solution in the moving frame is $U + h(t + t_\delta)$. The first time $\tau_{\delta,v}$ at which the orbital distance equals ρ_v satisfies

$$\lambda^{-1}\log\frac{\rho_v}{2\delta} \le \tau_{\delta,v} \le \lambda^{-1}\log\frac{\varepsilon_v}{\delta},$$

because $\|h(t+t_\delta)\|_X \le 2\delta e^{\lambda t}$ and the distance at $t = -t_\delta$ exceeds ρ_v. This proves (27). Formula (28) is the asymptotic in Theorem B.1 with $m = 4$. □

Remark 4.4 (Velocity dependence and the remaining assumption). The construction gives a fixed positive escape radius for each fixed nonzero small velocity. It does not bound that radius uniformly as $v \to 0$. The eigenvectors approach the neutral sector and the spectral projection constants may diverge. The rigorously implied rate statement along a unit direction e is the iterated limit

$$\lim_{r\downarrow 0} r\left[\lim_{\delta\downarrow 0}\frac{\tau_{\delta,re}}{\log(1/\delta)}\right] = \sqrt{\frac{\mathcal{D}_Q}{2e^T Te}}.$$

Critical scaling transfers the theorem to every frequency $\omega > 0$, with normalized velocity $v\omega^{-3/4}$ and the exact rate covariance $\lambda(\omega, v) = \omega\lambda(1, v\omega^{-3/4})$. The nonlinear instability mechanism is a classical Hamiltonian one; the present connection uses the previously computed critical small-velocity eigenvalue. Real-profile nondegeneracy is still an explicit hypothesis. The theorem does not assert finite-time blowup or classify the subsequent evolution after departure.

5 Conserved internal correlations and the critical mass

The quantitative polarization theorem has a consequence for every time of an actual solution. The internal correlation operator is conserved, and a field with more than one occupied spatial mode cannot approach the scalar sharp interpolation constant arbitrarily closely. This raises the mass range in which conservation of energy controls the whole evolution.

5.1 The common scalar flow

Let $\mathcal{H}$ be a nonzero separable complex Hilbert space and let a be a positive elliptic, real-even homogeneous symbol of order $m > 1$. Put $s = m/2$ and

$$\rho_U = \|U\|_{\mathcal{H}}^2, \qquad M(U) = \int \rho_U, \qquad A(U) = \int a(\xi)\|\widehat{U}(\xi)\|_{\mathcal{H}}^2\, d\xi.$$

As in the quantitative polarization theorem, take either $\mathcal{F}(U) = \int \rho_U^\kappa$, $\kappa = 1 + m/N$, or the positive definite critical Hartree interaction $\mathcal{F}(U) = \mathcal{D}(\rho_U, \rho_U)$, $\kappa = 2$. Write $r = \kappa - 1$. We allow a common real nonnegative multiplier b of order at most m, and write

$$iU_t = [a(D) + b(D)]U - V_{\rho_U}U, \qquad E_b(U) = \frac{A(U) + B_b(U)}{2} - \frac{\mathcal{F}(U)}{2\kappa}. \tag{37}$$

No homogeneity or dilation inequality on b is used in this section.

For the general flow statements we assume a conservative local $H^s(\mathbb{R}^N;\mathcal{H})$ theory with the continuation property: every finite endpoint of the maximal interval entails an unbounded H^s norm. A concrete quartic class satisfying this assumption is proved below. The sharp scalar mass M_c is fixed by

$$\mathcal{F}(u) \le \kappa(M(u)/M_c)^r A(u).$$

Define the positive trace-class operator on $\mathcal{H}$

$$\Gamma(U) = \int |U(x)\rangle\langle U(x)|\, dx, \qquad \mu(U) = \|\Gamma(U)\|_{\mathrm{op}}, \qquad \chi(U) = 1 - \frac{\mu(U)}{M(U)}. \tag{38}$$

For a nonzero field, $\operatorname{Tr}\Gamma = M$ and $0 \le \chi < 1$. The largest eigenvalue exists even if the internal space is infinite-dimensional, because Γ is compact.

Lemma 5.1 (Conservation of all internal correlations)**.** *Every energy-space solution of* (37) *satisfies* $\Gamma(U(t)) = \Gamma(U(0))$. *In particular,* μ *and* χ *are constant throughout its maximal interval.*

Proof. For fixed internal vectors h, k, the scalar components $u_h = \langle h, U\rangle$ and $u_k = \langle k, U\rangle$ solve the same equation $iu_t = H(t)u$, where $H(t) = a(D) + b(D) - V_{\rho_U}$ is a self-adjoint spatial form. The Hilbert-triple product rule for $CH^s \cap C^1H^{-s}$ functions gives

$$\frac{d}{dt}\int u_h\overline{u_k} = -i\langle H(t)u_h, u_k\rangle + i\langle u_h, H(t)u_k\rangle = 0.$$

The local nonlinear form is continuous by Hölder at exponent 2κ; the Hartree form is continuous by HLS. Equality for a countable dense set of h, k proves equality of the operators. The trace-norm estimate

$$\|\Gamma(U) - \Gamma(W)\|_1 \le (\|U\|_2 + \|W\|_2)\|U - W\|_2$$

also shows that this identity passes to energy-space limits. □

The conserved eigenvalues measure the masses in mutually orthogonal spatial modes after an internal unitary change of basis. Merely writing one spatial profile in many components still gives a rank-one operator and $\mu = M$.

5.2 A strict improvement retaining the scalar sharp mass

Theorem 5.2 (Internal mass distribution and energy control)**.** *There is a constant $c > 0$, depending only on the scalar dispersion and interaction and independent of the number of components, such that every nonzero $U \in H^s(\mathcal{H})$ satisfies*

$$\mathcal{F}(U) \le \kappa\left(\frac{M(U)}{M_c}\right)^r [1 - c\chi(U)]A(U). \tag{39}$$

One may take $c = \min\{C^{-1}, 1/2\}$, where C is the dimension-independent quantitative polarization constant.

Consequently, if the initial state satisfies

$$\eta := 1 - \left(\frac{M}{M_c}\right)^r (1 - c\chi) > 0, \tag{40}$$

its conservative energy-space evolution is global in both time directions, and

$$\sup_{t\in\mathbb{R}}\{A(U(t)) + B_b(U(t))\} \le \frac{2E_b(U(0))}{\eta}. \tag{41}$$

Thus every fixed $\chi > 0$ permits the mass interval

$$M < M_c(1 - c\chi)^{-1/r},$$

which extends strictly above the scalar critical mass.

Proof. Let $\mathfrak{d}$ and $\mathfrak{p}$ denote the normalized interpolation deficit and polarization distance:

$$\mathfrak{d}(U) = 1 - \frac{\mathcal{F}(U)}{\kappa(M(U)/M_c)^r A(U)}, \qquad \mathfrak{p}(U) = \inf_{\|\zeta\|=1, q\in H^s}\left\{\frac{M(U - \zeta q)}{M(U)} + \frac{A(U - \zeta q)}{A(U)}\right\}.$$

For fixed ζ, orthogonal projection gives

$$\inf_q M(U - \zeta q) = M - \langle\zeta, \Gamma(U)\zeta\rangle.$$

Taking the infimum over unit ζ and discarding the nonnegative kinetic term yields $\mathfrak{p}(U) \ge \chi(U)$. The quantitative polarization theorem gives $\mathfrak{p}(U) \le C\mathfrak{d}(U)$, proving (39).

For clarity, if $\Gamma_A = \int a(\xi)|\widehat{U}(\xi)\rangle\langle\widehat{U}(\xi)|\,d\xi$, the exact distance formula is

$$\mathfrak{p}(U) = 2 - \lambda_{\max}\left(\frac{\Gamma(U)}{M} + \frac{\Gamma_A}{A}\right).$$

The same scalar projection minimizes both terms in their combined Hilbert norm. Since Γ_A/A is positive with trace one, this also gives $\mathfrak{p} \geq \chi$. No conservation of Γ_A is asserted.

By Lemma 5.1, M, χ, and hence η are constant along the flow. Equation (39) gives

$$E_b(U(t)) \geq \frac{\eta}{2}A(U(t)) + \frac{1}{2}B_b(U(t)).$$

As $0 < \eta \leq 1$, conservation yields (41). Ellipticity, the conserved mass, and the continuation property finish the proof. □

Corollary 5.3 (Scalar reduction of critical-mass unbounded evolution)**.** *Suppose $M(U) = M_c$ and a conservative orbit has unbounded kinetic energy along a sequence of times, including possible approach to a finite maximal endpoint. Then $\Gamma(U(0))$ has rank one, and there is one fixed unit vector $\zeta \in \mathcal{H}$ such that*

$$U(t,x) = \zeta u(t,x)$$

throughout the maximal interval. Here u solves the corresponding scalar equation with mass M_c.

More generally, any energy-space-unbounded orbit of finite conserved mass satisfies

$$\chi \leq c^{-1}\left[1 - (M_c/M)^r\right]. \tag{42}$$

Proof. If $M = M_c$ and $\chi > 0$, Theorem 5.2 applies with $\eta = c\chi$, contradicting unboundedness. Thus $\mu = M$, so positivity and trace M force $\Gamma = M|\zeta\rangle\langle\zeta|$. For every $h \perp \zeta$,

$$\int |\langle h, U(t,x)\rangle|^2\,dx = \langle h, \Gamma h\rangle = 0.$$

A countable orthonormal basis in $\zeta^\perp$ shows that $U(t,x)$ lies in the same complex line almost everywhere. Substitution gives the scalar equation. Finally an unbounded orbit cannot satisfy (40); rearranging its negation proves (42). □

The scalar reduction holds for the initial datum and the entire orbit, not merely for a rescaled concentration limit. For $M = M_c(1+\varepsilon)$, (42) gives $\chi \leq (r/c)\varepsilon + O(\varepsilon^2)$.

5.3 A concrete quartic flow

Proposition 5.4 (Unconditional realization in dimensions one to three)**.** *Let $N = 1, 2, 3$, let a be any real-even positive elliptic homogeneous quartic symbol, and let b be real with $0 \leq b(\xi) \leq C(1 + |\xi|^4)$. Then*

$$iU_t = [a(D) + b(D)]U - \|U\|_{\mathcal{H}}^{8/N}U, \qquad U(0) \in H^2(\mathbb{R}^N; \mathcal{H}),$$

has a conservative local H^2 flow with the continuation property. Therefore Theorem 5.2 and Corollary 5.3 hold without additional flow assumptions for these equations, including countably many components.

Proof. The Fourier group is unitary on H^2. In these dimensions, H^2 embeds into L^∞, and the radial Hilbert-space map $z \mapsto \|z\|^{8/N}z$ is locally Lipschitz on $H^2(\mathcal{H})$. Indeed its first three derivatives are bounded on bounded balls because $8/N > 2$; the chain rule, the H^2 product estimates, and $H^1 \hookrightarrow L^4$ give the required estimate, with constants independent of internal dimension. Duhamel contraction gives the local flow and its continuation property. Symmetric Fourier Galerkin regularization gives mass and energy conservation; the regularized solutions converge in CH^2 on every local existence interval, and both conserved functionals are continuous there. Lemma 5.1 gives conservation of the full internal operator. □

Relation to kinetic-energy inequalities. Conservation of all component overlaps and the use of the largest occupation are established tools in nonlinear Schrödinger systems; see Hong–Kwon–Yoon, arXiv:1712.06811, Section 1 and Proposition 7.3. Their Theorem 2.3 also includes fractional kinetic-energy inequalities. The specific connection used here is the dimension-independent correction $1 - c(1 - \mu/M)$ to the *scalar sharp* critical-mass inequality. This gives a strict extension above M_c for every genuinely nonpolarized internal state and the exact initial-data reduction at mass M_c. The constant c is not claimed to be optimal, and equality in (40) is not covered.

6 Relation to prior work and the remaining evolution problem

Boulenger–Lenzmann [1], Theorem 3, prove finite- or infinite-time blowup for radial negative-energy data in the pure critical quartic equation, in dimensions at least two. Bonheure–Castéras–Gou–Jeanjean [2], Theorem 1.2, deduce the corresponding instability of radial ground states, without real-sector nondegeneracy. The recent Dinh–Muse manuscript [3], Theorem 1.1, states finite-time blowup for radial negative-energy data, while its Remark 1.4 leaves the nonradial problem open. Theorem 2.1 treats unrestricted spatial geometry but adds zero momentum and small mass excess, and concludes quantitative amplification rather than blowup. It is not a strengthening of the radial finite-time conclusion.

Instability obtained with radial perturbations already implies instability in the full perturbation space of that same radial wave. Moreover, Lenzmann–Sok [4], Theorem 2, prove radial symmetry of all sharp optimizers when the Lebesgue exponent is even. In the quartic critical problem this includes $N = 1, 2, 4$; the radial instability results above therefore already cover all ground states for $N = 2, 4$. The distinct assertion here is the quantitative constraint on arbitrary nonradial near-threshold zero-momentum trajectories. The targeted comparison found no matching $d^{5/3}$ estimate, but does not establish exhaustive priority. Gou's cylindrical result [5] imposes both cylindrical symmetry and an axial moment condition and does not give that unrestricted statement.

Existence of traveling ground waves for general dispersion is studied by Bugiera–Lenzmann–Schikorra–Sok [6]. We include the small-velocity ground-wave compactness proof needed for constrained descent. Hamiltonian unstable subspaces and group bounds are established tools; Lin–Zeng [7] provide a general trichotomy theory. Our nonlinear construction uses these ideas in a concrete form to turn the computed critical small-velocity eigenvalue into an ancient orbit and an exit time. Real L_+ nondegeneracy remains an assumption for that result, and is not supplied by the universally valid phase gap.

Conservation of mutual component inner products is established in unitary nonlinear Schrödinger systems, including Hong–Kwon–Yoon [8], Proposition 7.3. Their generalized kinetic-energy inequalities also include fractional orders and the relevant endpoint. We do not claim those tools as new. The connection used here retains the sharp scalar M_c, introduces a quantitative loss $1 - c\chi$ through polarization, and then uses the conserved χ to control every time of the evolution. The resulting enlarged boundedness interval is sufficient; it is not claimed to be the exact mixed-state critical mass.

What is still missing. A lower bound on the largest kinetic norm does not exclude a global bounded solution that reaches that large norm. Excluding such a trajectory, and then distinguishing finite-time concentration from infinite-time growth, remain separate unresolved steps. Zero momentum does not set the fourth-order mass-transport velocity to zero: that velocity involves a cubic Fourier moment. A sharper long-time link between conserved momentum, core motion, and changes of scale is the next central issue. We have not proved that link here.

A Sharp density geometry and quantitative polarization

This appendix reproduces the analytic inputs used by the new evolution arguments. They are proved here, including the infinite-dimensional compactness step. No radiality, positivity of the kinetic semigroup, or real-sector nondegeneracy is used.

Let $\mathcal{H}$ be any nonzero separable complex Hilbert space. Finite-dimensional $\mathbb{C}^d$ and ℓ^2 are both allowed. Fix $N \geq 1$, $m > 1$, $s = m/2$, and a real-even, positive elliptic, m-homogeneous symbol a, of class C^{N+2} away from zero. For $U \in H^s(\mathbb{R}^N; \mathcal{H})$ put

$$\rho_U = \|U\|_{\mathcal{H}}^2, \quad M(U) = \int \rho_U, \quad A(U) = \int a(\xi)\|\widehat{U}(\xi)\|_{\mathcal{H}}^2 \, d\xi.$$

We use either of the following interactions:

$$\begin{aligned} &\kappa = 1 + m/N, && \mathcal{F}(U) = \int \rho_U^\kappa, && V_\rho = \rho^{\kappa-1}; \\ &\kappa = 2, && \mathcal{F}(U) = \mathcal{D}(\rho_U, \rho_U), && V_\rho = K * \rho. \end{aligned}$$

In the second case $1 < m < N$, the real even kernel K is homogeneous of degree $-m$, its interaction form $\mathcal{D}$ is positive definite, and

$$|K(x)| + |x||\nabla K(x)| \leq C|x|^{-m}.$$

As before, positivity of the Fourier transform of K almost everywhere suffices for positive definiteness. Pointwise positivity is not needed. The scalar sharp inequality defines M_c by

$$\mathcal{F}(u) \leq \kappa \left(\frac{M(u)}{M_c}\right)^{\kappa-1} A(u). \tag{43}$$

Theorem A.1 (Sharp interpolation and dimension-independent compactness)**.** *Inequality* (43) *has the same sharp constant for $\mathcal{H}$-valued functions. Every optimizer has the form*

$$U(x) = \zeta Q(x), \qquad \|\zeta\|_{\mathcal{H}} = 1,$$

where Q is a real scalar optimizer, up to a constant phase. If $M(U_n) \to M_c$, $A(U_n) \to 1$, and $\mathcal{F}(U_n) \to \kappa$, then the sequence is strongly compact in H^s after translations and internal unitaries. This remains true when the internal dimensions vary: embed all spaces in ℓ^2 before applying the unitaries.

A.1 Sharp interpolation and complete polarization

The local interaction is the κ-th power of the L^κ density norm. The Hartree interaction is the square of its positive definite Hilbert density norm. Thus both have the form $\mathcal{F}(U) = \|\rho_U\|_{\mathcal{B}}^\kappa$ for a strictly convex real density space $\mathcal{B}$. In an internal orthonormal basis $U = (u_j)$, the triangle inequality, scalar sharp interpolation, and sequence Hölder give

$$\begin{aligned} \mathcal{F}(U)^{1/\kappa} &\leq \sum_j \mathcal{F}(u_j)^{1/\kappa} \\ &\leq \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} \sum_j A(u_j)^{1/\kappa} M(u_j)^{(\kappa-1)/\kappa} \\ &\leq \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} A(U)^{1/\kappa} M(U)^{(\kappa-1)/\kappa}. \end{aligned}$$

Finite partial sums converge in the density norm because the same bound applied to their tails tends to zero. A scalar optimizer placed in one component proves sharpness.

Split the components further into real scalar functions f_j. Equality forces every active f_j to be a scalar optimizer, all f_j^2 to be proportional, and all kinetic-to-mass ratios to agree. The same

conclusions hold after any real orthogonal rotation of a pair f_i, f_j. If f_j/f_i took both signs of a nonzero constant, the squared sum and difference could not be proportional; when the constant has magnitude one their nonzero supports would be disjoint. Thus each f_j is one real constant multiple of a fixed scalar function. This proves complete polarization, including for countably many components.

A.2 Compactness modulo internal unitaries

Ordinary local compactness does not hold directly in an infinite internal dimension: a fixed spatial shape can move through mutually orthogonal internal vectors. The following argument removes precisely that loss of compactness.

Lemma A.2 (Internal-unitary local compactness)**.** *Let $X = H^s(\mathbb{R}^N;\mathbb{C})$ with $s > 0$. From every bounded sequence in $X\widehat{\otimes}\ell^2$, one can extract a subsequence and internal unitaries W_n such that W_nU_n converges weakly in $X\widehat{\otimes}\ell^2$ and strongly in $L^2_{\mathrm{loc}}(\mathbb{R}^N;\ell^2)$.*

Proof. Fix an orthonormal basis (ϕ_k) of the scalar Hilbert space X. Write its internal coefficient vectors as $c_{nk} = \langle \phi_k, U_n\rangle_X \in \ell^2$. Gram–Schmidt the first n vectors and extend the resulting finite orthonormal list to an internal orthonormal basis. A unitary W_n can therefore be chosen so that

$$W_n c_{nk} \in \operatorname{span}\{e_1,\dots,e_k\}, \qquad k \le n.$$

For each fixed k, these vectors have a convergent subsequence in a fixed finite-dimensional space. Diagonal extraction gives strong convergence of every coefficient vector. Weak compactness gives a weak limit of W_nU_n, with exactly those coefficients.

For $\chi \in C_c^\infty$, the scalar multiplication map $K_\chi : X \to L^2$, $f \mapsto \chi f$, is compact. If P_L is the scalar projection onto the first L basis vectors, then

$$\|K_\chi(I - P_L)\|_{X\to L^2} \longrightarrow 0.$$

Tensoring with the identity on ℓ^2 preserves this norm. The transformed P_LU_n converge strongly because there are only L coefficient vectors; after multiplication by χ, the tails are uniformly small. This proves local strong convergence. □

Compactness assertion of Theorem A.1. The localized Sobolev inequality has constants independent of internal dimension: use the Hilbert-valued Bessel-potential representation and its norm triangle inequality, then spatial localization and the sum of component quadratic norms. Thus local mass vanishing for an H^s-bounded sequence implies vanishing in $L^p(\mathcal{H})$ for $2 < p < 2N/(N-2s)$, or every finite $p > 2$ if $2s \ge N$. In the local case take $p = 2\kappa$; in the Hartree case use $p = 4N/(2N-m)$ and Hardy–Littlewood–Sobolev. Hence $\mathcal{F}(U_n) \to \kappa > 0$ excludes local mass vanishing.

Translate a nonzero amount of local mass into a fixed ball and use Lemma A.2. We obtain a nonzero weak limit U, strong local L^2 convergence, and by interpolation strong local convergence in the interaction exponent. Put $R_n = U_n - U$. The quadratic quantities split. For local powers the Brezis–Lieb argument for $\|U\|_{\mathcal{H}}^{2\kappa}$ gives

$$\mathcal{F}(U_n) = \mathcal{F}(U) + \mathcal{F}(R_n) + o(1).$$

For Hartree, the term $\operatorname{Re}\langle U, R_n\rangle$ tends to zero in $L^{p/2}$ by local convergence and the small L^p tail of U. Also $\mathcal{D}(\rho_U, \rho_{R_n}) \to 0$: approximate U in $L^p(\mathcal{H})$ by a compactly supported smooth finite-dimensional field. Its potential is bounded and decays at infinity, while R_n tends locally to zero in L^2. HLS controls the approximation error. This proves the same splitting for Hartree.

Let $M_1 > 0, M_2$ and A_1, A_2 be the limiting contributions. Then $M_1 + M_2 = M_c$, $A_1 + A_2 = 1$, and sharp interpolation yields

$$M_c^{\kappa-1} \le M_1^{\kappa-1}A_1 + M_2^{\kappa-1}A_2.$$

Both masses cannot be positive, since each would be strictly below M_c. Thus $M_2 = 0$, and the inequality then forces $A_2 = 0$. The convergence is strong in H^s. The limit is an optimizer, already proved to be completely polarized. □

A.3 Phase coercivity and the internal symmetry kernel

For completeness, the scalar phase fact used below follows directly from the density norm. Normalize a real optimizer by $M(Q) = M_c$, $A(Q) = 1$, and put $\omega_* = (\kappa - 1)/M_c$. The form

$$L_Q = a(D) + \omega_* - V_{Q^2}$$

is nonnegative: differentiate the sharp inequality at $Q + i\sqrt{t}\, h$. If h belongs to its kernel, write $x = A(h)$ and $y = M(h)/M_c$. The supporting-functional inequality for the density norm, sharp interpolation for h, and weighted arithmetic–geometric mean give

$$x + (\kappa - 1)y \le \kappa x^{1/\kappa} y^{(\kappa-1)/\kappa} \le x + (\kappa - 1)y.$$

Equality implies $x = y$ and $h^2 = yQ^2$ by strict convexity. The kernel of a nonnegative form is linear; apply the same conclusion to $h + Q$ to get $h = cQ$. The potential is compact as a form $H^s \to H^{-s}$, by local Rellich and Hölder in the local case, or the compact-potential approximation and HLS in the Hartree case. Consequently

$$\langle h, L_Q h\rangle \ge c_Q \|h\|_{H^s}^2, \qquad \langle h, Q\rangle_{L^2} = 0. \tag{44}$$

This holds for complex h by real–imaginary splitting, and it can be summed over arbitrarily many components with the same constant.

At Qe_1, write a perturbation as $(h + ik)e_1 + Z$ with $Z \perp e_1$. The Hessian of $E + \omega_* M/2$ is

$$\langle h, L_+ h\rangle + \langle k, L_Q k\rangle + \sum_{j\ge 2} \big(\langle \operatorname{Re} Z_j, L_Q \operatorname{Re} Z_j\rangle + \langle \operatorname{Im} Z_j, L_Q \operatorname{Im} Z_j\rangle\big).$$

Its kernel in the phase and internal transverse sector is exactly

$$\{Q\eta :\ \eta \in \mathcal{H}, \operatorname{Re}\langle \eta, e_1\rangle = 0\}.$$

For $\mathcal{H} = \mathbb{C}^d$, this is the $(2d-1)$-dimensional real tangent space to the internal unitary orbit. Any additional full Hessian kernel lies in the scalar real sector L_+; no assumption on that sector is made.

A.4 Quantitative polarization without real-sector nondegeneracy

The preceding compactness and phase coercivity also give a quantitative statement for arbitrary fields, not just stationary or traveling ones. Define the scale-invariant interpolation deficit

$$\mathfrak{d}(U) = 1 - \frac{\mathcal{F}(U)}{\kappa (M(U)/M_c)^{\kappa-1} A(U)}$$

and the squared distance to a completely polarized field by

$$\mathfrak{p}(U) = \inf_{\substack{\|\zeta\|_{\mathcal{H}}=1 \\ q\in H^s}} \left\{ \frac{M(U - \zeta q)}{M(U)} + \frac{A(U - \zeta q)}{A(U)} \right\}.$$

Both are defined for every nonzero $U \in H^s(\mathcal{H})$.

Theorem A.3 (Dimension-independent quantitative polarization)**.** *There is $C < \infty$, depending only on the fixed scalar dispersion and interaction, such that*

$$\mathfrak{p}(U) \le C\mathfrak{d}(U) \tag{45}$$

for every nonzero Hilbert-valued field U and every internal dimension. No nondegeneracy or uniqueness of the scalar real-sector ground state is required.

Proof. Amplitude and spatial scaling preserve both quantities, so impose $M(U) = M_c$, $A(U) = 1$. Write $\|U\|_X^2 = M(U)/M_c + A(U)$. Since $\mathfrak{p}(U) \le 2$, failure of a uniform constant would produce U_n with $\mathfrak{d}(U_n) \to 0$ and $\mathfrak{d}(U_n)/\mathfrak{p}(U_n) \to 0$. After the symmetries of Theorem A.1, $U_n \to Qe_1$ strongly in H^s. Align $\int QU_n$ with the positive real first internal direction and write $U_n = (u_n, Z_n)$. Each component of Z_n is complex L^2-orthogonal to Q, and $u_n \to Q$, $Z_n \to 0$ in H^s.

The interaction expansion in these transverse directions is

$$\mathcal{F}(u_n, Z_n) = \mathcal{F}(u_n) + \kappa\langle V_{|u_n|^2}, \|Z_n\|^2\rangle + o(\|Z_n\|_X^2). \tag{46}$$

For local powers, set $a = |u_n|^2$ and $b = \|Z_n\|^2$ pointwise. When $1 < \kappa \le 2$, the nonnegative Taylor remainder of $(a+b)^\kappa$ after its linear term in b is at most $C_\kappa b^\kappa$. When $\kappa > 2$, it is at most $C_\kappa(a^{\kappa-2}b^2 + b^\kappa)$. Hölder and the dimension-independent $L^{2\kappa}$ Sobolev bound make these respectively $O(\|Z_n\|_X^{2\kappa})$ and $O(\|Z_n\|_X^4 + \|Z_n\|_X^{2\kappa})$, since u_n stays bounded. Both are $o(\|Z_n\|_X^2)$. For Hartree the remainder is exactly $\mathcal{D}(\|Z_n\|^2, \|Z_n\|^2) = O(\|Z_n\|_X^4)$.

Put $m_n = M(Z_n)$ and $a_n = A(Z_n)$. Scalar sharp interpolation gives

$$\mathcal{F}(u_n) \le \kappa(1 - m_n/M_c)^{\kappa-1}(1 - a_n).$$

Combining its expansion with (46) and potential form convergence yields

$$\begin{aligned}
\kappa - \mathcal{F}(U_n) &\ge \kappa\left[a_n + \frac{\kappa-1}{M_c}m_n - \langle V_{Q^2}, \|Z_n\|^2\rangle\right] - o(\|Z_n\|_X^2) \\
&= \kappa\sum_j \langle Z_{nj}, L_Q Z_{nj}\rangle - o(\|Z_n\|_X^2) \\
&\ge c\|Z_n\|_X^2.
\end{aligned}$$

The constant is independent of component number. Taking the candidate $e_1 u_n$ in the distance gives $\mathfrak{p}(U_n) \le \|Z_n\|_X^2$. As $\mathfrak{d}(U_n) = (\kappa - \mathcal{F}(U_n))/\kappa$, this contradicts the assumed ratio and proves the theorem. □

The theorem controls internal polarization rather than the full scalar shape. It therefore remains valid even when scalar optimizer uniqueness or real-sector quantitative stability is unavailable.

B The spectral pair created by slow translation

The velocity–momentum identities determine the inertia of the charge Hessian. We now identify the pair of spectral values that emerges from the critical scaling mode, including its leading coefficient. The result assumes real-profile nondegeneracy; this hypothesis is not supplied by phase coercivity alone.

Let a be positive, elliptic, real-even and homogeneous of degree $m > 1$. Consider

$$iu_t = a(D)u - |u|^{2m/N}u, \qquad a(D)U + \omega U + iv\cdot\nabla U - |U|^{2m/N}U = 0.$$

All Hilbert spaces in this section are real unless explicitly complexified. Let $Q = U_{1,0}$ be a real sharp ground wave and set

$$\Lambda Q = \frac{N}{2}Q + x\cdot\nabla Q, \qquad L_- = a(D) + 1 - |Q|^{2m/N}, \qquad L_+ = a(D) + 1 - \left(1 + \frac{2m}{N}\right)|Q|^{2m/N}.$$

We impose the following hypotheses.

1. The functions $\partial_j Q$ and ΛQ have the regularity and spatial decay needed to differentiate translations and critical scaling in L^2 and in the profile equation. In particular, $L_+\Lambda Q = -mQ$ holds in the operator sense.

2. $L_- \geq 0$, $\ker L_- = \mathbb{R}Q$, and L_-^{-1} is bounded on $Q^\perp$. Moreover $\ker L_+ = \operatorname{span}\{\partial_j Q : 1 \leq j \leq N\}$, and L_+ has a bounded inverse on that kernel's orthogonal complement.
3. There is a C^1 local profile branch $U_{\omega,v}$, closed under critical scaling up to phase and translation. Its reduced action $d(\omega, v) = E(U_{\omega,v}) + \omega M(U_{\omega,v})/2 - v \cdot P(U_{\omega,v})/2$ is C^2 and satisfies $d_\omega = M/2$ and $d_v = -P/2$. Profile differentiation is valid, and in a translation gauge

$$D_{v_j} U_{1,0} = -iL_-^{-1}\partial_j Q.$$

4. On the real space $\mathcal{H} = L^2(\mathbb{R}^N; \mathbb{C})$ the real action Hessians $\mathcal{L}_v$ are selfadjoint on a common dense domain. Their kernels are exactly

$$K_v = \operatorname{span}_{\mathbb{R}}\{iU_{1,v}, \partial_1 U_{1,v}, \ldots, \partial_N U_{1,v}\}.$$

   These kernel bases vary continuously in $\mathcal{H}$. The inverse of $\mathcal{L}_v$ on $K_v^\perp$, extended by zero on K_v, exists and is norm-continuous as a bounded operator on $\mathcal{H}$ for small v.

The last hypothesis explicitly includes a uniform complementary spectral gap. Common-domain operator continuity, or suitable norm-resolvent continuity with an isolated kernel of fixed dimension, provides standard ways to verify it. It is not being inferred for every dispersion symbol.

Define the real matrix and vector

$$T_{ij} = \langle \partial_i Q, L_-^{-1}\partial_j Q\rangle, \qquad c_j = \langle \Lambda Q, L_-^{-1}\partial_j Q\rangle,$$

and the scalar

$$\mathcal{D}_Q = \langle \Lambda Q, L_-^{-1}\Lambda Q\rangle - c^T T^{-1} c. \tag{47}$$

The inverse of L_- always acts on $Q^\perp$.

Theorem B.1 (Small-velocity spectral asymptotics)**.** *Under the four hypotheses above, T is positive definite and $\mathcal{D}_Q > 0$. If $m \neq 2$, then for every sufficiently small nonzero v, the nonzero spectrum of*

$$\mathcal{A}_v = J\mathcal{L}_v, \qquad J = -i,$$

in a fixed neighborhood of zero consists of exactly one simple pair $\lambda_\pm(v)$, satisfying

$$\lambda_\pm(v)^2 = \frac{(m-2)v^T T v}{\mathcal{D}_Q} + o(|v|^2). \tag{48}$$

For $m > 2$ this pair is real and

$$\lambda_\pm(v) = \pm\sqrt{\frac{(m-2)v^T T v}{\mathcal{D}_Q}} + o(|v|).$$

In particular, the wave is spectrally unstable. For $1 < m < 2$ the pair is purely imaginary, with

$$\lambda_\pm(v) = \pm i\sqrt{\frac{(2-m)v^T T v}{\mathcal{D}_Q}} + o(|v|).$$

The latter assertion concerns the pair near zero and does not, by itself, exclude unstable spectrum elsewhere.

Proof. Phase coercivity gives $T > 0$: a nonzero directional derivative of a nonzero L^2 function cannot vanish. Put

$$\Lambda_\perp Q = \Lambda Q - \sum_j (T^{-1}c)_j \partial_j Q.$$

Then

$$\mathcal{D}_Q = \langle \Lambda_\perp Q, L_-^{-1}\Lambda_\perp Q\rangle \geq 0.$$

Equality would imply $(x-\zeta)\cdot\nabla Q + NQ/2 = 0$ for some ζ. Thus Q would be fixed by the mass-preserving dilation group about ζ. A nonzero function homogeneous of degree $-N/2$ cannot belong to L^2: its radial mass integral is proportional to $\int_0^\infty dr/r$. Hence $\mathcal{D}_Q > 0$. The same Schur-complement formula shows that $\mathcal{D}_Q$ is unchanged if a linear combination of translations is added to ΛQ, so it is independent of the spatial center.

Write $U_v = U_{1,v}$ and $g_0 = iU_v$, $g_j = \partial_j U_v$. Their span K_v is isotropic:

$$\langle g_i, Jg_j\rangle = 0. \tag{49}$$

For the phase–translation pairings this follows from integration of a mass derivative. For two translations it follows from commutation of translation charges, or directly from the Fourier representation of $\langle \partial_i U, -i\partial_j U\rangle_{\mathbb{R}}$.

Let Π_v project orthogonally onto K_v, let $\Pi_c = I - \Pi_v$, and write $\mathcal{L}_c = \mathcal{L}_v|_{K_v^\perp}$ and $J_c = \Pi_c J \Pi_c$. The operator

$$R_v(\lambda) = (\mathcal{L}_c + \lambda J_c)^{-1}$$

is extended by zero on K_v. A Neumann series and hypothesis 4 show that this inverse is analytic for $|\lambda| < \rho$, with a common $\rho > 0$ for small v. The eigenvalue problem is considered on the complexification of $\mathcal{H}$; real pairings below extend bilinearly.

The equation $J\mathcal{L}_v\psi = \lambda\psi$ is equivalent to $(\mathcal{L}_v + \lambda J)\psi = 0$. Decompose $\psi = k + w$ with $k \in K_v$ and $w \in K_v^\perp$. For $\lambda \neq 0$ the complementary equation gives

$$w = -\lambda R_v(\lambda) J k.$$

By (49), the kernel equation is therefore exactly $D(\lambda, v)z = 0$, where

$$D_{ij}(\lambda, v) = \langle g_i, JR_v(\lambda)Jg_j\rangle, \qquad k = \sum_{j=0}^{N} z_j g_j. \tag{50}$$

The common factor suppressed in this equation is $-\lambda^2$. For $0 < |\lambda| < \rho$, analytic block elimination therefore preserves the eigenvalues and their algebraic multiplicities. Selfadjointness of $\mathcal{L}_c$ and skew-adjointness of J imply

$$D(-\lambda, v) = D(\lambda, v)^T. \tag{51}$$

Profile differentiation yields

$$\mathcal{L}_v \partial_\omega U = -U, \qquad \mathcal{L}_v \partial_{v_j} U = -i\partial_j U = Jg_j.$$

Components of the parameter derivatives in K_v do not contribute, by (49). Consequently

$$D(0, v) = Cd''(1, v)C, \qquad C = \mathrm{diag}(1, -I_N).$$

The lower-right block is $D_{vv}(0, v) = -B(1, v)/2$, where $B = D_v P$. The first derivative of the branch gives

$$P(U_v) = 2Tv + o(|v|), \qquad B(1, v) = 2T + o(1).$$

Thus the lower-right block stays invertible for small λ, v. Take its scalar Schur complement $f(\lambda, v)$. Identity (51) makes f even in λ. It is consequently analytic in $z = \lambda^2$ in a common disk, with coefficients continuous in v.

The exact scaling identity for the reduced action gives

$$\begin{aligned} f(0, v) &= \frac{P^T B^{-1} P - (m-1)v\cdot P}{2m^2} \\ &= \frac{2-m}{m^2} v^T T v + o(|v|^2). \end{aligned} \tag{52}$$

For completeness, the identity follows by differentiating the scaling laws for the charges. With $H = d''$ and $D_* = (m\omega, (m-1)v)^T$, they give $HD_* = (0, -P/2)^T$ and $H_{vv} = -B/2$. Eliminating the velocity block shows that its scalar Schur complement is $[P^T B^{-1} P - (m-1)v \cdot P]/(2m^2\omega^2)$. This uses no differentiated second-order remainder for the profile.

It remains to compute the coefficient of λ^2 at $v = 0$. Let R_0 be the L^2-orthogonal projection of $\Lambda Q/m$ off the translation kernel, and put $Z_j = L_-^{-1}\partial_j Q$. At $v = 0$ the two real blocks give

$$\mathcal{L}_0 = \operatorname{diag}(L_+, L_-), \qquad J(h, b) = (b, -h),$$

$$\mathcal{L}_c^{-1} J g_0 = (-R_0, 0), \qquad \mathcal{L}_c^{-1} J g_j = (0, -Z_j).$$

Expanding the exact resolvent in (50), and writing $b_j = \langle R_0, Z_j \rangle$, gives

$$\begin{aligned} D_{00}(\lambda, 0) &= \lambda^2 \langle R_0, L_-^{-1} R_0 \rangle + O(\lambda^4), \\ D_{0v}(\lambda, 0) &= \lambda b^T + O(\lambda^2), \\ D_{v0}(\lambda, 0) &= -\lambda b + O(\lambda^2), \\ D_{vv}(\lambda, 0) &= -T + O(\lambda). \end{aligned}$$

The scalar Schur complement is even. Its coefficient is therefore

$$\partial_z f(0, 0) = \langle R_0, L_-^{-1} R_0 \rangle - b^T T^{-1} b = \frac{\mathcal{D}_Q}{m^2} > 0.$$

The final equality uses invariance of the Schur expression under addition of translations to the scaling derivative. Analyticity with a common radius and continuity in v now give

$$f(\lambda, v) = \frac{2-m}{m^2} v^T T v + o(|v|^2) + \lambda^2 \left(\frac{\mathcal{D}_Q}{m^2} + o(1) \right) + O(\lambda^4),$$

where the last remainder is uniform for small v.

The analytic scalar equation, regarded as an equation in $z = \lambda^2$, has one simple real root near zero, by $\partial_z f(0, 0) > 0$ and uniform analytic inversion. That root obeys (48). Its sign is the sign of $m - 2$, since $T > 0$, and it is nonzero for nonzero sufficiently small v. Its two square roots are simple because $\partial_\lambda f = 2\lambda \partial_z f \neq 0$ there. The invertible complementary and velocity blocks exclude further nonzero spectrum in a sufficiently small fixed neighborhood. □

Remark B.2 (Center, frequency, and linear time scale). If $Q(-x) = Q(x)$, then $c = 0$ by parity. Without this symmetry, omitting $c^T T^{-1} c$ gives an incorrect coefficient in general. Critical scaling also gives the exact spectral covariance

$$\lambda(\omega, v) = \omega \lambda\big(1, v\omega^{-(m-1)/m}\big).$$

Thus for $m > 2$ and fixed $\omega > 0$,

$$\lambda_+(\omega, v) = \omega^{1/m} \sqrt{\frac{(m-2) v^T T v}{\mathcal{D}_Q}} + o(|v|).$$

The reciprocal growth rate is of order $|v|^{-1}$. This appendix alone concerns linear exponential growth; Section 4 proves a nonlinear exit time in the specified quartic class. When $m = 2$, the displayed leading coefficient vanishes; a general homogeneous degree-two symbol need not be a quadratic form, so an exact Galilean degeneracy is not asserted.

Corollary B.3 (Critical Hartree interactions). *Theorem B.1 holds, with the same coefficient, for the critical Hartree flow*

$$i u_t = a(D) u - (K * |u|^2) u, \qquad K(x) = |x|^{-m}, \quad 1 < m < N,$$

whenever its real ground wave and local branch satisfy the same four hypotheses with

$$L_- = a(D) + 1 - (K * Q^2), \qquad L_+ h = L_- h - 2Q\big(K * (Qh)\big).$$

It also holds for a positive-type anisotropic interaction of the same homogeneity under those operator and regularity assumptions.

Proof. The proof above uses the nonlinearity through phase and translation invariance, conjugation symmetry, critical scaling, and the stated real-block operators. These properties hold for the Hartree energy. In particular $L_+ \Lambda Q = -mQ$, the parameter equations and the charge-scaling identities are unchanged. No step in the resolvent reduction uses locality of L_+. Thus the proof applies verbatim. □

Remark B.4 (Relation to established instability theory). The general mechanism of a spectral pair leaving zero at charge Hessian degeneracy belongs to established Hamiltonian bifurcation theory. Here the additional calculation is its coefficient for mass-critical translation branches, including the coupling to the translation modes. The theorem proves spectral instability for $m > 2$ under its explicit assumptions. Nonlinear orbital instability requires a sufficiently differentiable local flow and the associated linear evolution bounds or unstable-manifold construction in a specified phase space; it is not inferred here for every admissible symbol and dimension. Likewise, real-profile nondegeneracy is not proved by this section.

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